Cheap Signals, Costly Proof: The Reach and Limits of Award-Layer Screening in Cartel Enforcement

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Abstract

Cartel enforcement must allocate costly proof-producing effort before proof exists. Using São Paulo's BEC procurement platform and CADE adjudications, we ask how far routine award records can rank where bid-level forensics should begin. We study a minimal award-layer construct—persistent zero-win participation, implemented as a frequent-loser flag—validated against a reproducible, adjudication-anchored cobidder label that never uses the screen. Validating an administrative screen against adjudicated cases without adjusting for procurement opportunity systematically over-credits it: decomposed by opportunity, timing, and case concentration, the cheap signal's apparent reach mostly reflects mechanical co-participation exposure and case concentration, the residual ordering net of opportunity is marginal at best, and prospective ranking fails outside incumbent pools. The contribution is the organizational result—the award-to-bid recovery decision is a sequential cost-recall problem whose frontier, not any cutoff, is the design object—and a disciplined audit protocol. Liability remains in the richer evidentiary record.

Keywords: cartel screening, public procurement, award-layer data, enforcement triage, incomplete observability, bid rigging

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1. Introduction

Cartel enforcement begins before legal proof exists. An agency deciding where to investigate rarely starts with the communications, bid histories, leniency testimony, or tender-level conduct evidence that would support liability. It starts with thinner information: administrative records, complaints, procurement anomalies, repeated interactions, and case leads that make some firms and markets worth a closer look. The first legal-economic problem is therefore not proof itself. It is the allocation of proof-producing effort. The Becker–Stigler tradition treats investigation and sanctions as costly instruments whose intensity must be chosen optimally (Becker, 1968; Stigler, 1970); the margin we study is the one prior to any sanction—how a resource-constrained authority sequences costly evidence acquisition before it knows whom to charge. This is the enforcement-side counterpart to the procurement-governance questions in Decarolis et al. (2020), where bureaucratic competence and discretion shape award outcomes: there the question is how a buyer runs a tender well; here it is how an enforcer decides where to spend forensic capacity once the tenders are run.

That distinction matters in procurement. The empirical cartel-screening literature usually starts from the bid layer: submitted bid amounts, ranks, dispersion, histories, and sometimes bidder networks. Classic screens compare bidding under competition and collusion (Porter and Zona, 1993; Bajari and Ye, 2003); variance and descriptive screens formalize suspicious bid distributions (Abrantes-Metz et al., 2006; Harrington, 2008) and field-test them in procurement (Imhof et al., 2018; Imhof, 2019); recent work extends the approach through bidder groups (Conley and Decarolis, 2016), machine learning (Huber and Imhof, 2019; Wallimann et al., 2023), and robust screen design (Chassang et al., 2022). Those objects are close to the evidentiary stage but not always cheap at scale. This is the central asymmetry the paper exploits: many procurement systems make contract-award records routinely visible—who participated, who won, item codes, and negotiated prices—while keeping per-bidder bid files in operational systems that must be recovered, cleaned, and inspected case by case. The award layer is cheap and broad but legally thin; the bid layer is forensically rich but costly to open.

Cheap administrative red flags built from exactly such routine records are already in institutional use. Corruption-risk indicators such as single bidding and repeat-participation anomalies are computed from contract-award data across entire procurement systems and used by multilateral institutions and audit bodies precisely because bid-level records are costly to recover (Fazekas and Kocsis, 2020; OECD, 2022). The zero-win construct we study is the cartel-facing member of this cheap-red-flag family: a statistic an authority can run over award data before committing to forensic recovery. The paper audits the reach of this family before it steers scarce forensic effort. An enforcement agency thus faces a sequencing problem—before opening the costly bid layer, can it use the cheap award layer to decide where bid-layer forensics should begin?

This paper studies that question on São Paulo's BEC procurement platform, covering 1,654,401 tender-items from 2009 to 2019. BEC records a routine award layer—participants, winners, item codes, negotiated prices, procuring units, years, and modalities—and also preserves a richer bid layer recoverable through the LANCES export. CADE procurement-cartel adjudications provide external legal anchors. This combination lets us study an earlier stage than most bid-distribution screens: the point at which an agency observes award records but has not yet recovered the bid-level evidence needed for forensic evaluation.

The award-layer statistic is deliberately simple. Among firms with zero wins, we rank participation intensity using $\log(1 + \text{tenders\_count})$ and implement an auditable binary rule that flags *frequent losers*. The logic is monotone rather than classificatory: if coordinated bidders use loser-side participation to create the appearance of competition while avoiding the win, persistent zero-win participation can rank loser-side exposure under a transparent behavioral condition. The statistic does not identify a firm type or assign cartel membership; the rank allocates forensic attention, while the richer bid layer remains where tender-level conduct and agreement are evaluated.

That legal scope determines the validation target. Direct CADE defendants are legal anchors, but they are not the natural object of a zero-win loser-side screen. Procurement cartels often allocate winning roles, rotate designated winners, and use other firms on the losing side of the tender (Pesendorfer, 2000; Asker, 2010); the same role-allocation logic is central in cartel theory and case evidence (Marshall and Marx, 2012), and large-scale procurement evidence shows how such arrangements appear in administrative bidding records (Kawai and Nakabayashi, 2022). We therefore validate the ranking against reproducible adjudication-anchored exposure rather than against cartel membership: the target is the set of 651 unique always-loser firms that share at least one tender-item with a BEC-active direct CADE defendant, with the defendants themselves excluded. The label is rebuilt end-to-end from the current pipeline scripts, and—this point matters for the validity of every comparison that follows—the frequent-loser flag never enters its construction. Direct defendants anchor legally adjudicated environments; the always-loser cobidders are the loser-side footprint the award layer can plausibly rank before bid-level proof is recovered.

The empirical design is organized around three threats that would each, on its own, make such a ranking uninformative. The first is opportunity arithmetic: high-volume zero-win firms mechanically meet more defendants, so a raw co-bidding association may reflect participation volume and opportunity-set exposure rather than any behavioral signal. The second is timing: a ranking formed using contemporaneous or later participation cannot order forensic priority prospectively. The third is case concentration: a single adjudicated case can manufacture apparent platform-wide reach. We discipline the ranking against all three—opportunity-cell adjustment, strict rolling-origin timing, and leave-one-case-out tests—alongside a bid-layer benchmark; Section 4 and the appendices report the battery, and Appendix C gives the exact exposure, leakage, and timing audit construction.

The results are deflationary, and that is the point. Read raw, the award-layer ranking does concentrate adjudication-anchored cobidders inside the always-loser stratum; the decomposition then shows how little survives discipline. Most of the raw concentration reflects procurement opportunity, because firms that participate more, in the same environments as adjudicated defendants, are mechanically more exposed to them. Within comparable opportunity sets the residual ordering is marginal at best and not robust across designs: held to matched cells the ranking is close to chance, the increment over an exposure-only model is small, and matched-stratum permutation and enrichment tests do not reject the opportunity-only null. Strict prospective ranking fails outside the incumbent pool, and the apparent reach is case-sensitive—a single adjudicated case supplies a large share of the positives and of the firms a fixed-budget queue would open. The ranking is also silent about direct CADE defendants, consistent with its loser-side scope rather than any cartel-membership reading.

Where the framework earns its keep is in cost accounting and forensic sequencing. On the same adjudication-anchored target the award-layer ranking is comparable to a transparent bid-moment random-forest benchmark inspired by Imhof–Wallimann-style screens, and the two combine only conditionally—a combined model improves precision under random cross-validation but falls below the award-only ranking under case-grouped folds. Used as a gatekeeper, the award layer traces a cost-recall frontier: across survivor-pool sizes it retains much of the adjudication-anchored recall of full bid-layer scoring while shrinking the bid-recovery footprint. The reduction is real but must be read against its denominator—large in firms opened, far smaller in tender-items and bid rows, because the survivors are high-participation firms—and there is no optimal cutoff: the frontier itself, not any single operating point, is the enforcement-design object. Because strict timing for the bid rerank is not available with the current bid-layer features, this is a retrospective cost-footprint design conditional on the validated incumbent ranking, not a fully prospective deployment test.

This paper makes three contributions. The first is organizational, and it is the law-economics-organization payload: the award-to-bid recovery decision is a sequential evidence-allocation problem, and the object an enforcement designer chooses is not any cutoff on the cheap screen but the cost-recall frontier that the screen traces when used as a gatekeeper. Reframing procurement-cartel screening this way turns it from a classification exercise—does the flag attain a high AUC—into a sequencing problem: how a resource-constrained authority orders costly evidence acquisition under incomplete observability, where suspicion is cheap and proof is dear (Becker, 1968; Stigler, 1970; Harrington, 2008; Sánchez Graells, 2019; Chassang et al., 2022). Existing bid-distribution screens evaluate suspicious bidding once bid-level data have been recovered; we study the prior stage, where agencies observe award records but must decide where recovery should start, and we show that the frontier—not a detector-versus-detector horse race—is the design margin.

The second contribution is a disciplined audit protocol whose components are individually standard but whose sequencing for the award-to-bid decision is, to our knowledge, new: a non-circular adjudication-anchored label, opportunity-cell adjustment, strict rolling-origin timing, leave-one-case-out tests, a bid-layer benchmark, and cost-recall accounting, applied to ask not whether a cheap screen attains a high pooled AUC but how much of that AUC is genuine ranking signal rather than mechanical opportunity exposure, retrospective information, or concentration in a single adjudicated case. We do not claim a validated, transferable framework; what is portable is the *principle*—validating an administrative screen against adjudicated cases without adjusting for procurement opportunity systematically over-credits it—and the order in which the audit steps must be run.

Third, applying that protocol to São Paulo's BEC yields a map of reach and limits for this setting: persistent zero-win participation concentrates a legally limited, adjudication-anchored exposure target in raw terms, but the residual ordering net of opportunity is marginal at best, prospective ranking fails outside incumbent pools, and the apparent reach is concentrated in one case—so the estimated ranking is not a portable cartel score, while the audit principle and the cost-recall trade-offs it exposes are what transfer. Locating these boundaries is itself the enforcement-design product.

Two scope points round out the design. Price evidence enters only as scope information, not as a damages exercise. And strategic response shapes implementation: a static cutoff invites gaming, while refreshed thresholds, continuous ranks or capacity-constrained queues, bunching checks, and bid-layer follow-up fit the staged enforcement architecture.

The broader lesson is institutional. A legal system that waits for full proof before ranking investigative priorities will spend scarce forensic capacity poorly; a system that treats cheap suspicion as proof will overreach. The useful middle ground is a disciplined triage rule—cheap enough to run before proof exists, informative enough to order forensic attention within a known reach, and legally limited enough to leave liability where it belongs, in the richer evidentiary record. Knowing where such a rule stops ranking is as much a part of disciplined enforcement as knowing where it begins.

2. Costly Proof and Observable Awards

The paper's empirical object is an institutional sequence, not a stand-alone classifier. An enforcement agency first observes a thin administrative record, then decides whether to spend resources recovering richer bid-level evidence, and only later can that evidence contribute to legal proof. This section defines that sequence for BEC procurement: the award and bid information layers, and the adjudication-anchored labels used to validate a loser-side ranking. The legal predicate is what fixes the screen's role. Proof of a procurement cartel requires evidence of coordinated conduct—communications, allocation rules, bid rotation, cover bidding, or tender-level patterns inconsistent with independent rivalry. Brazilian competition law treats bid rigging as an infraction under Lei 12.529/2011; comparable frameworks likewise require evidence beyond suspicious administrative patterns (Hovenkamp, 2005; Baker, 2003; Wils, 2007, 2016). A pattern in award records does not contain those legal predicates, so an award-layer screen produces *forensic priority*, not an accusation: its output orders where costly bid recovery should start, and liability stays in the bid layer.

2.1. Award and Bid Layers in BEC

São Paulo's BEC platform preserves the two information layers an enforcement sequence requires. The sample contains 1,654,401 tender-items from 2009 to 2019, dominated by two modalities: *Convite*, a sealed-bid invitation procedure under Lei 8.666/1993 with a statutory three-bidder quorum and a low value cap; and *Pregão*, a reverse electronic auction under Lei 10.520/2002 with real-time offers and no minimum-bidder rule.[2] Both generate a routine award record before any case-specific bid recovery occurs. Auction format, bureaucratic incentives, discretion, and rules can shape both bidding behavior and observed outcomes (Athey et al., 2011; Decarolis, 2014; Coviello and Gagliarducci, 2017; Decarolis et al., 2020, 2025).

The *award layer* is that routine administrative envelope. It records participant identity,

[2]Throughout, a firm's participation is counted once per tender-item it enters. In Pregão, a bidder may revise its offer many times within a single tender-item; these iterative offers do not increase the participation count T_i, which is defined at the tender-item level.

winner identity, item code, negotiated price, procuring unit, year, and modality. These fields show who appeared, who won, what was bought, and where a firm repeatedly participated without winning. They do not reveal the bids submitted by each participant, the timing of offers, bid revisions, or within-tender dispersion. The award layer is therefore cheap enough for broad monitoring, but too thin to supply legal proof of coordination.

The *bid layer* is the forensic-recoverable layer. BEC preserves per-bidder offers, timestamps, and bid-revision sequences in the LANCES export. Those records support bid-distribution forensics: within-tender dispersion, skewness, spread, distance from the winning bid, and related transparent bid-moment features inspired by Imhof–Wallimann-style screens (Imhof et al., 2018; Imhof, 2019), including machine-learning variants that combine these bid moments at scale (Huber and Imhof, 2019; Wallimann et al., 2023). But these features require the bid file to be recovered, cleaned, linked, and inspected; they are not available at the same cost or timing as an award-record query. The bid layer is thus available but costly—feasible for case work and validation, not the natural first object for system-wide triage.

Figure 1 summarizes the two layers. Bid-distribution methods remain the natural forensic tools once bid microdata have been recovered; award-layer triage asks whether the thinner routine record can rank where that recovery should start.

2.2. CADE Anchors and Adjudication-Anchored Exposure Labels

CADE adjudications supply the legal anchors, not the screen's labels mechanically. The linked BEC portfolio contains 12 adjudicated procurement-cartel cases with 65 firm-defendants, of which 47 are active in BEC during the sample window. Final decisions span 2015–2025, with long process-to-ruling lags. That timing is central: whatever value a cheap ranking has lies in moving investigative attention before the completed adjudication record exists—a premise this paper audits rather than assumes.

The adjudication record creates two objects with different roles. Direct CADE defendants are the legal anchors: firms named in adjudicated procurement-cartel cases and matched to BEC participant records. They locate environments where legal process established cartel conduct, but they are not the natural target of a zero-win loser-side

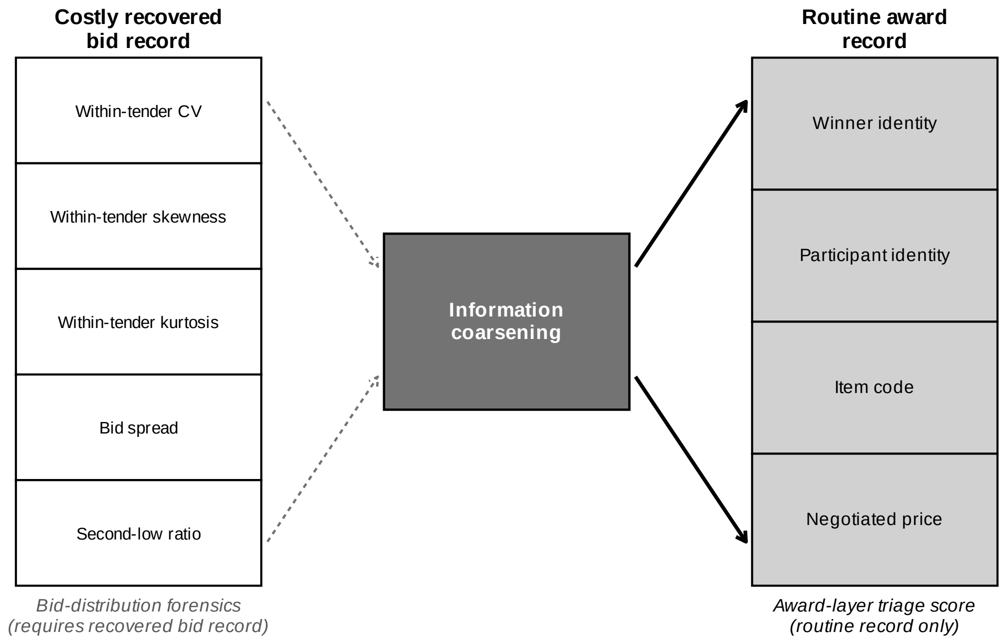

Figure 1: Information layers in procurement-cartel enforcement. This is an information-cost diagram, not a detector horse race. Bid-distribution forensics requires per-bidder bid amounts and within-tender bid moments; the award-layer screen uses participant identity, winner identity, item code, and negotiated price. The bid layer is not absent: it is costly to recover, clean, and inspect from the LANCES export. The question is whether the cheap award layer can rank where the costly bid layer should be opened first. A pattern in award records does not imply legal proof of coordination.

Alt text: Two vertical columns of stacked boxes represent two information layers. The left column lists costly recovered bid-record features (within-tender bid moments such as dispersion, skewness, spread, and second-low ratio). The right column lists routine award-record fields (winner identity, participant identity, item code, negotiated price). A central gate links them: the thinner award layer is available earlier and more cheaply and is used to rank where the costly bid layer should be recovered first.

statistic—defendants may occupy winner-side, allocation, or rotating roles, and a screen built from repeated losing should not be judged as if it were a generic membership classifier over adjudicated defendants.

The validation target is narrower. A cobidder is a unique always-loser firm that shares at least one BEC tender-item with a BEC-active direct CADE defendant, yielding 651 always-loser cobidders. The frequent-loser flag is not used to construct the label; of the 651 cobidders, 341 are frequent losers and 310 are not. Cobidder status is not legal membership or a latent truth about agreement. It is an adjudication-anchored exposure label: the firm is observed on the losing side near a legally established cartel anchor, appropriate for evaluating forensic priority, not liability. Collapsing defendants and cobidders into a single cartel class would overstate what award records establish; requiring a loser-side screen to recover direct defendants would misstate what the statistic is designed to rank. The relevant question is whether persistent zero-win participation concentrates the loser-side firms most plausibly worth bid-layer follow-up around adjudicated cartel environments—a question the validation design in Section 4 disciplines against opportunity, timing, and case concentration.

Table 1 fixes the roles used throughout the paper: the outer BEC population, the always-loser stratum the award layer can rank, the frequent-loser operational screen, the direct-defendant legal anchors, and the cobidder loser-side validation target.

The rest of the paper keeps these objects separate: the screen operates inside the always-loser stratum, the validation target is cobidder adjacency, direct defendants anchor the environments without defining the screen's success, and bid-layer forensics remains the proof-producing stage. This separation lets the paper evaluate a legally limited triage architecture without turning suspicion into adjudication.

3. Award-Layer Triage

The award layer observes outcomes before it observes conduct. A losing bid may be sincere, poorly priced, capacity constrained, exploratory, strategic, or coordinated. The screen therefore uses what the award layer can measure at the triage stage: which firms appeared, which won, and which kept appearing without ever winning. The object is a

Table 1: Data Layers, Populations, and Legal-Economic Roles

Object	Size / layer	Role in the analysis	Legal interpretation
Award layer	Routine record	Participants, winners, item codes, prices, procuring units, years, and modalities; the cheap triage layer.	Administrative record; not proof.
Bid layer	LANCES export	Per-bidder offers, timestamps, revisions, and bid moments; the costly forensic layer.	Forensic record; can support bid-level analysis.
All BEC participants	41,444 firms	Outer firm universe for award-record construction.	Outer universe.
Always-losers	16,843 firms	Zero-win stratum in which the award-layer screen ranks loser-side exposure.	Screening stratum.
Frequent losers	2,735 firms	Operational first-stage flag within always-losers; not a legal category.	Operational flag; not a legal category.
Direct CADE defendants in BEC	47 firms	Legal anchors from adjudicated procurement-cartel cases.	Legal anchor.
Always-loser cobidders	651 firms	Always-losers sharing a tender-item with direct defendants; main validation target.	Exposure label; not membership.

Notes: Direct CADE defendants are legal anchors. Always-loser cobidders are adjudication-anchored exposure labels. Frequent losers are an operational screen, not a legal category. The categories separate legal membership from screening exposure. Benchmark-specific cobidder counts and the funnel reconciliation appear in Table 2.

ranking for forensic priority, not proof of conduct or a classification of cartel membership. This section defines that ranking, the operational queue it yields, and the implications that discipline it.

3.1. Persistent Zero-Win Participation

Let T_i denote the number of BEC tender-items in which firm i participates, and let W_i denote the number it wins. In Pregão, a firm may revise its offer repeatedly within one tender-item; T_i counts tender-item participations, so these iterative offers do not raise T_i.The award-layer screen operates inside the zero-win stratum, $\{i : W_i = 0\}$, which contains 16,843 firms. Within that stratum, the continuous score is

$$s_i = \log(1 + T_i).$$

The log transformation stabilizes the scale but does not change the ordering. The economic object is the rank: among firms the award layer observes only as losers, which ones appear persistently enough to warrant bid-layer follow-up?

The maintained condition is monotonic rather than classificatory. If cartel-deployed losing participants are present, they should appear more persistently in the zero-win stratum than ordinary losing firms, because such a role is useful only through repeated deployment in cartel-targeted tenders. The condition is maintained, not derived; Appendix B formalizes it as a monotone-likelihood-ratio ranking result. The main text uses it more modestly: persistent zero-win participation is a cheap statistic that can rank loser-side exposure under a transparent behavioral assumption and direct attention to the bid layer where conduct can be evaluated. It orders firms for forensic priority; it does not assign a firm type.

The zero-win restriction is a scope restriction: many firms never win for ordinary competitive reasons (poor fit, high costs, capacity constraints, market exploration, subcontracting positioning, or bad luck), and the screen's content comes from persistence inside that heterogeneous loser-side set. Appendix B adds the dynamic margin behind this: ordinary zero-win firms face participation costs and may exit after repeated losses, while a losing role that carries strategic value or compensation can sustain persistence, and a platform-survival audit finds that frequent-loser cobidders remain active markedly longer than other frequent losers—consistent with persistence as a triage signal, not a legal category. If the ranking has enforcement value, it should concentrate adjudication-anchored loser-side adjacency beyond what generic participation volume, product-market overlap, and opportunity to meet CADE anchors would mechanically produce.

3.2. From Ranking to an Operational Queue

The continuous score is the central object because agencies can translate a rank into different investigative protocols: inspect the top k firms under fixed capacity, refresh the rank each quarter and examine bunching below old cutoffs, or request LANCES bid files for a survivor pool before applying bid-distribution forensics. All preserve the same sequencing logic: award records first, costly bid recovery second.

For an auditable binary implementation, we also report a frequent-loser flag marking zero-win firms whose participation count exceeds the median plus 1.5 times the interquartile range of the always-loser participation distribution. In BEC, the statistical cutoff is 13.5, implemented as $T_i \geq 14$, which flags 2,735 firms, or 16.2% of the always-loser stratum. The cutoff is fixed from the participation distribution alone, before any CADE label enters the analysis.

The $T_i \geq 14$ rule is an auditable administrative implementation of the rank, not a structural or legal threshold.The paper does not rest on the number 14 but on the continuous loser-side participation ranking. Two analysts applying the same rule to the same award records obtain the same first-stage list; above the cutoff does not mean cartel member, and below it does not mean clean. Because a published static threshold can be gamed, the enforcement architecture treats the binary flag as one implementation among several—continuous ranks, top-k queues, refreshed thresholds, and budget-calibrated survivor pools—and the strategic-adaptation analysis later returns to this point.

3.3. Testable Implications

The ranking can organize incumbent-firm triage only if it survives the threats implied by its limited information. The core implication is the validation claim: persistent zero-win participation should rank adjudication-anchored loser-side cobidders within the always-loser stratum, beyond what mechanical exposure—high participation volume and operation in the same procurement environments as CADE defendants—would produce under ordinary competition. The remaining implications are scope (the statistic should perform worse against direct defendants than cobidders, ranking a different role), economic content of the cobidder target, non-redundancy of the award layer relative to bid-distribution forensics, gatekeeping value at affordable budgets, and vulnerability of the published binary threshold to strategic adaptation. These testable implications, and the threats that would weaken each, are evaluated in Section 4 and the appendix; the screen is assessed at the stage for which it is designed—the allocation of costly forensic attention before liability evidence is assembled. The BEC cutoff is an implementation detail; the portable object is the staged architecture that uses cheap administrative records to discipline where costly cartel forensics should begin.

4. Validating the Loser-Side Ranking

This section validates the award-layer ranking, not the legal status of ranked firms. The validation target is the one defined in Section 2.2: always-loser firms observed alongside direct CADE defendants in adjudicated procurement-cartel environments. These cobidders carry an adjudication-anchored exposure label; they are not legal members of a cartel. A useful ranking must do more than report a high AUC. The main threat is exposure—firms that participate often, or operate in the same environments as defendants, have more opportunities to appear near those anchors even under ordinary competition, so a screen must concentrate loser-side firms beyond what participation volume and opportunity alone imply (Porter and Zona, 1993; Bajari and Ye, 2003; Conley and Decarolis, 2016; Kawai and Nakabayashi, 2022). Two further threats are timing (the screen should rank before the target information is used) and legal overreach (it should not be read as a generic cartel-membership detector).

Subsection 4.1 constructs the label and reconciles the validation samples. Subsection 4.2 is the central exercise: it disciplines the ranking against procurement opportunity and asks what, if anything, the score adds net of exposure. Subsection 4.3 imposes strict timing and case-composition discipline. Subsection 4.4 folds the direct-defendant scope check into a bounded reading of what the validation establishes.

4.1. Label Construction and Sample Reconciliation

The full CADE legal portfolio comprises 12 adjudicated procurement-cartel cases with 65 legal firm-defendants. We match defendants to BEC participant records by exact 14-digit CNPJ equality, with no fuzzy or manual step; 47 are active in BEC and define the direct-defendant set. Their defendant tender-items (52013 distinct items) locate the adjudicated environments.

The main validation label is constructed from current scripts by matching BEC-active direct CADE defendants to tender-items and identifying unique always-loser firms that share at least one BEC tender-item with those anchors. Direct defendants are excluded. The frequent-loser flag is not used to define the label; it is the award-layer score evaluated against the label. This yields 651 always-loser cobidders, of which 341 are frequent losers and 310 are not—the label spans both sides of the screen's threshold, so the validation cannot be circular in the score. The conservative benchmark restricts the CADE portfolio to the 4 cases judged on or before 2020 under the *same* definition, yielding 208 cobidders from 19 matched defendants; it is a strict subset of the main label. Table 2 reports the funnel. Positive counts differ across later tables only because later exercises impose common-support, timing, or bid-feature restrictions on the candidate pool; Table A.3 in the appendix reconciles each denominator.

Table 2: Label-Construction Funnel and Benchmark Reconciliation

Sample / benchmark	Cases	Legal def.	BEC-active def.	AL cobid.	FL among	Cobidder def.	Role
Full CADE portfolio	12	65	47	651	341	shared tender-item	reference
Main validation target	12	65	41	651	341	shared tender-item	primary
Conservative pre-2020	4	–	19	208	107	shared tender-item (same)	early-cases subset

Notes: A cobidder is an adjudication-anchored exposure label, not membership. The main label is a unique always-loser firm (win rate = 0 over 2009–2019) sharing at least one BEC tender-item with a BEC-active direct CADE defendant; direct defendants are excluded; the frequent-loser flag is not used to construct the label. "FL among" is the descriptive composition of positives, not a label restriction. The 47 cited active defendants correspond to 48 distinct crossmatch CNPJ, of which 41 are present in the firm–tender map. Counts are unique firms.

Before adding stricter discipline, a baseline question: do firms flagged by the award-layer rule appear disproportionately among always-loser firms near adjudicated anchors?

They do. A frequent loser is a cobidder with probability 12.5% (341/2,735), against 2.2% for non-frequent-loser always-losers (310/14,108)—an enrichment ratio near 5.7×. On the full always-loser pool the continuous score ranks the label at ROC-AUC 0.761 (PR-AUC 0.143; precision@500 0.216, lift 5.6×), and the binary flag at 0.688. Persistent zero-win participation is not random with respect to adjudication-anchored loser-side exposure. But this baseline is not yet evidence of anything beyond arithmetic, because participation volume and shared environments mechanically manufacture proximity. The next subsection asks how much of the ranking survives once opportunity is held comparable—and the answer is: very little.

4.2. Opportunity-Adjusted Validation

This is the central validation exercise. The threat is mechanical: a firm with high participation T_i co-appears with defendants more often by construction, and a firm in the same item, buyer, year, and modality cells is more exposed even under ordinary competition. We measure opportunity directly. For each always-loser firm we count observed contact with direct defendants, O_i, and construct expected contact under random matching, $E_i = \sum_g n_{ig}\, p_g$, where n_{ig} is participation in cell g and p_g its defendant-density. We build E_i at three cell granularities and restrict to common support; the exposed subsample retains $n = 6{,}040$ always-loser firms containing all 651 positives (construction in Appendix C.1). Holding E_i fixed, we ask whether the score reproduces the label better than exposure alone, whether it adds anything net of exposure, and whether it ranks within equal-opportunity strata.

The answer is deflationary on every margin, and we are explicit about which comparisons carry the weight. Exposure-based benchmarks must be read in tiers, because contact-derived quantities partially encode a contact-defined label. At the mechanical extreme, ranking by observed contact O_i reproduces the label at ROC-AUC 0.905 and a plug-in E_i reaches 0.985—these encode the label (a cobidder *is* a firm with $O_i > 0$) and are reported only to expose the inflation, as are the combined exposure-plus-score logits near 0.99 (Table 3). The firm-level leave-one-out E_i used as the exposure benchmark reaches 0.855; an audit of that benchmark shows it, too, is partially label-encoding—recomputing each cell's defendant density from *non-cobidder* rows only, so that no eventual positive contributes to any firm's expected contact, leaves a label-blind opportunity ranking at only 0.553 (Appendix C). Genuine opportunity structure, purged of the label, therefore explains little on its own. What the tiers establish is not that "opportunity is the better model" but something sharper: *any* contact-derived benchmark—and any contact-defined label—is mechanically entangled with participation volume, which is exactly why raw validation over-credits the screen.

Second, conditional on opportunity the residual ordering is marginal at best and not robust across designs—and the within-stratum design itself survives an audit of its ability to detect. The nested model that adds the score to the exposure-only model raises AUC by 0.010 (DeLong $p = 0.013$; DeLong et al., 1988)—small and only marginally significant. *Within* opportunity strata, where firms are compared only against others with comparable expected contact, the continuous score ranks the label at AUC 0.471 and the binary flag at 0.507—both indistinguishable from chance. This null is falsifiable, not built in: within the *same* strata a positive-control score (O_i itself) ranks the label at 0.953, so the design detects a residual when one exists; and the within-stratum AUC is stable across cell granularities (0.508–0.600 from coarse to strict cells; Appendix C). Because label-soaked strata could in principle flatten a true signal, we also stratify on the *label-blind* expected contact: there the score retains 0.722—but the negative controls in Section 5 show this same ordering arises under placebo anchors and under non-CADE high-volume-winner anchors, identifying it as generic co-participation geometry rather than anything cartel-specific. A coarsened-exact opportunity match yields AUC 0.626, and the within-stratum gap in cobidder probability between flagged and non-flagged always-losers is −0.017—negative. The score does rank positive excess contact $O_i - E_i > 0$ at AUC 0.764, but that diagnostic shares the score's participation component and cannot carry the residual claim alone.

The exposure-preserving permutation tests settle the question. Under a pure-exposure binomial null the observed PR-AUC 0.143 does not beat the null (null mean 0.264; empirical $p = 1.00$): a pure-exposure model out-predicts the raw score on the rare-target metric. And under the permutation that shuffles cobidder labels only within matched exposure-by-participation strata—the design that isolates any residual signal—the observed PR-AUC does *not* reject the null either ($p = 0.127$), and the frequent-loser enrichment within matched strata is likewise not significant ($p = 0.067$). These non-rejections are informative, not merely underpowered: a calibration that injects synthetic residual signal into the labels shows the permutation test is correctly sized under the null and rejects with probability 0.97 once the true within-stratum AUC reaches 0.55 (probability 1.00 at 0.60), so the non-rejection bounds any surviving residual well below the weakest ordering that could matter operationally (Appendix C). Both permutation distributions are reported there. Nor is this an artifact of label breadth: restricting positives to firms sharing at least two defendant tender-items strengthens the conclusion rather than reversing it—the exposure-only model improves faster than the score, the within-stratum AUC stays at chance (0.506), and the nested increment vanishes (+0.003, $p = 0.47$; Appendix C).

The reading we carry forward is therefore stronger than caution: under a reproducible, non-circular label, the award-layer score does not reproduce adjudication-anchored exposure any better than procurement opportunity does, and once opportunity is held fixed there is no robust residual ordering left. The cheap signal's apparent reach is opportunity arithmetic. What survives for practice is the audit itself: an agency that runs this decomposition learns, before opening any bid record, that loser-side concentration should not be used as a stand-alone prioritization device in this setting—and the same audit, run on another platform, would show where (if anywhere) a cheap screen retains residual value there. The remaining threats—timing, leakage, and case composition—sharpen the same conclusion and are taken up next.

4.3. Timing, Leakage, and Case Composition

The previous subsection found no robust residual ordering net of opportunity. This subsection asks the operational question anyway —can the raw ranking, whatever its source, be used prospectively?—and the answer compounds the downgrade. The ranking is a retrospective, incumbent-pool diagnostic, concentrated in a few adjudicated cases, and it does not support prospective, platform-wide deployment.

Table 3: Opportunity-Adjusted Validation of the Award-Layer Ranking

Design	Score	N	Pos.	ROC-AUC	PR-AUC	Prec@500	Rec@500	95% CI / p
Raw score	log_tc	16,843	651	0.761	0.143	0.216	0.166	$[0.741, 0.780]$
Raw FL14	fl14	16,843	651	0.688	0.097	0.128	0.098	$[0.669, 0.707]$
Exposure-only benchmark	$\log E_i$	6,040	651	0.713	0.300	0.372	0.286	$[0.691, 0.735]$
Within-opportunity stratum	log_tc | opp	6,040	651	0.471	—	—	—	pooled C (chance)
Within-opportunity stratum	fl14 | opp	6,040	651	0.507	—	—	—	pooled C (chance)
Nested increment over exposure	+log_tc	6,040	651	+0.010	—	—	—	DeLong $p = 0.013$
Score ranks excess contact	log_tc	16,843	475	0.764	0.095	0.150	0.158	$1[O_i - E_i > 0]$
Permutation B (pure exposure null)	log_tc	16,843	651	—	0.143	—	—	$p = 1.00$
Permutation C (matched strata)	log_tc	16,843	651	—	0.143	—	—	$p = 0.127$
Matched / stratified (CEM-opp)	log_tc	6,040	651	0.626	—	—	—	$\Delta = -0.017$

Notes: The validation target is adjudication-anchored exposure (cobidder status) among always-loser firms, not cartel membership; the label never uses the frequent-loser flag (341 of 651 positives are frequent losers, 310 are not). Expected contact $E_i = \sum_g n_{ig} p_g$ is built from procurement opportunity cells (item, buyer, year, modality); cell definitions and retained support are in Appendix C.1. PR-AUC is the primary metric because the target is rare (651/16,843). The exposure-only benchmark reproduces the label better than the raw score on the rare-target metric; within-stratum and matched designs are at or below chance; the matched-strata permutation does not reject. Combined exposure-plus-score logits (ROC-AUC near 0.99) are deliberately omitted because E_i mechanically encodes the label and their fit is not score evidence.

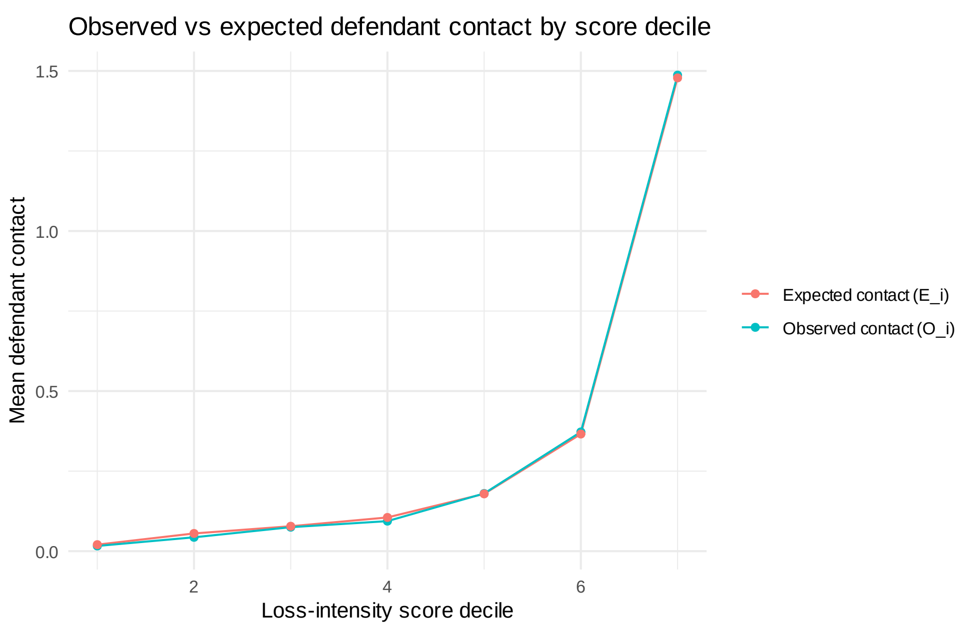

Figure 2: Opportunity-adjusted diagnostic: observed defendant contact O_i versus expected contact E_i under random matching, by award-layer score bin. Expected contact is built from procurement opportunity cells (item, buyer, year, modality); the comparison shows how much of the apparent score–contact relation survives the opportunity benchmark. *Alt text:* A binned plot with award-layer score bins on the horizontal axis and defendant-contact counts on the vertical axis. Two series are shown per bin, observed contact and expected contact under random matching; the two series track each other closely across score bins, indicating that expected contact under random matching accounts for most of the observed gradient.

A retrospective rank measures; it does not deploy. Every benchmark anchors its labels to CADE adjudications that close years after the bidding they describe, so even a high in-sample AUC would certify construct validity, not field performance. The strict-timing design freezes the score, the always-loser status, and the frequent-loser threshold to the 2009–2016 window and evaluates against 2017–2019 labels. On the realistic full candidate universe—every firm scorable at the screening date, with entrants carried at score zero—discrimination is at or below chance: ROC-AUC 0.474, PR-AUC 0.014, and **precision@500** $= 0$ **and recall@500** $= 0$. A screen that returns zero true positives in its top 500 candidates cannot be deployed. Inside the training always-loser pool the frozen score retains ROC-AUC 0.684 (binary flag 0.646), which is the incumbent-triage residue: a retrospective ordering among firms the agency already observes. The mechanical cause of the collapse is structural: of the 651 positives, 153 (23.5%) are entrants with no pre-window history, so the frozen score cannot rank them; only 498 positives are rankable at all, and roughly a quarter of the universe is tied at score zero. A loser-side score built from past losing cannot order firms that have not yet lost. The rolling-origin design confirms this: on the full universe every test year from 2014 to 2019 returns precision@500 and recall@500 of zero, with ROC-AUC between 0.446 (2015, below chance) and 0.50. A fully label-frozen variant closes the residual leakage concern in this design: freezing the pool, the score, *and* the label ingredients to the training window (always-loser status from 2009–2016 wins only; defendant contact from 2009–2016 tender-items only) leaves $13,051$ rankable incumbents, among whom the frozen score ranks *new* defendant contact arising in 2017–2019 at AUC 0.713 (231 positives; fully frozen retrospective contact: 0.718). Read against the negative controls of Section 5, this is participation volume forecasting generic future contact with active firms—not a cartel-specific signal that the stricter designs missed—and the full-universe zeros above remain the operative deployment result. The rolling-origin curves and the label-frozen construction are in Appendix C.

Case composition is the second structural fact. The largest case (trens/metros, CADE process 08700.004617/2013-41) supplies 32.0% of all positives and 45.4% of the top-500 true positives. Dropping it cuts PR-AUC from 0.143 to 0.090 (-37%) and precision@500 from 0.216 to 0.132; dropping the top two cases leaves PR-AUC 0.058.

ROC barely moves (0.761 → 0.763), which is precisely why ROC is the wrong lead metric here—it is insensitive to the case concentration the operational metrics expose. Case-balanced precision@500 is 0.043 against the firm-weighted pooled 0.216; six linkable cases contribute top-500 true positives, the largest accounting for 45–50% across cutoffs. Concentration also appears across environments: the largest item-group holds 21.5% of positives and 85.4% of positives sit in Pregão. A clustered randomization-inference test (item-group×year cells, $B = 1{,}000$) shows the pooled ordering is distinguishable from a within-cell shuffle ($p = 0.001$) but case-coverage breadth is not ($p = 0.103$): what the raw ranking picks up is concentrated exposure, not breadth. The leave-one-case-out distribution is reported in Appendix C; Table 4 synthesizes the battery.

The most this design supports is incumbent-firm triage under retrospective adjudication-anchored validation: an ordering among firms the agency already observes, useful as one input to where a forensic team looks first, and explicitly not a prospective, platform-wide screen. The estimated ranking is not a portable cartel score. The transferable object is the decomposition framework: label construction, opportunity adjustment, timing discipline, case-composition audit, bid-layer comparison, and cost-recall accounting.

Table 4: Timing and Case-Composition Synthesis

Design	Information set	N	Pos.	ROC-AUC	PR-AUC	Prec@500	Rec@500	Interpretation
Full-sample retrospective diagnostic	Full retrospective	16,843	651	0.761	0.143	0.216	0.166	Construct validity only; upper bound
Strict full test universe	Strict; entrants at score 0	41,444	651	0.474	0.014	0	0	Collapse: no prospective ranking
Strict rankable (incumbent)	Strict; prior-history firms	32,682	498	0.471	0.013	0	0	Entrant exclusion does not rescue
Strict training always-loser pool	Strict; train-window AL	21,819	651	0.684	0.077	0.124	0.095	Incumbent-triage residue
Rolling-origin (2014–2019)	Strict; full universe	per-year	381–651	0.45–0.50	≈ 0.012	0	0	Zero at the top every year
Leave-largest-case-out	Full retrospective	16,393	443	0.763	0.090	0.132	0.149	−37% PR-AUC
Leave-top-two-cases-out	Full retrospective	16,361	268	0.752	0.058	0.092	0.172	Thin residual tail
Clustered randomization inference	Label shuffle in cells	16,843	651	0.761	0.143	0.216	—	$p = 0.001$; coverage $p = 0.103$
Direct-defendant scope	Strict and full scores	—	47	0.49–0.70	—	—	—	Scope boundary (Section 4.4)

Notes: PR-AUC, precision@500, and recall@500 are the primary metrics because the target is rare; ROC-AUC is reported alongside but is insensitive to the case concentration the operational metrics expose. The full-sample retrospective row is a diagnostic *upper bound*, not a timing claim. In the strict and rolling-origin designs the score is frozen before the test window; later CADE adjudications are still used as labels but were not legally observable at the screening date, so these designs distinguish prospective scoring from retrospective adjudication. Cobidders are adjudication-anchored exposure labels, not cartel members. The strict full universe carries entrants at score zero; roughly a quarter of that universe is tied at zero. "0" denotes an exact zero true positives in the top 500.

4.4. Direct-Defendant Scope and Bounded Reading

The direct-defendant exercise aligns the validation target with the screen's legal-economic role. Direct CADE defendants are the legal anchors that locate adjudicated environments; they need not be persistent zero-win firms, since defendants may occupy winner-side, allocation, or rotating roles. The award-layer flag ranks the loser side by construction, and against direct defendants the binary flag is uninformative: AUC 0.491, indistinguishable from random. Continuous participation volume, by contrast, ranks defendants moderately above chance (AUC 0.66–0.70 across designs)—defendants are active firms, so volume alone carries some defendant-side information even though the loser-side flag does not. The role separation in the data is consistent with this: 14.9% of direct defendants are always-losers, and their median win rate is 0.261, whereas the cobidder target sits in the zero-win stratum by construction. The lesson is scope, not detection: a loser-side flag is silent about the winner side of adjudicated conduct, and nothing in this paper ranks defendants. The full scope matrix is in Appendix C.

Taken together, the benchmarks draw a boundary rather than certify a screen. Each addresses a distinct pre-registered threat—legal anchoring, participation volume, market exposure, timing, leakage, and scope asymmetry—and the consolidated threat map is in Appendix C. Persistent zero-win participation concentrates adjudication-anchored loser-side exposure in the raw data, but the decomposition shows that this concentration is what procurement opportunity already implies: the residual ordering net of opportunity is marginal at best and not robust across designs, the prospective ranking fails outside incumbent pools, and the raw signal is concentrated in a few adjudicated cases. The defensible product is the audit, not the screen: a reproducible decomposition that tells an enforcement agency how far a cheap award-layer statistic can be trusted before costly bid-layer work begins—here, not far—and that transfers to other platforms where the answer may differ. The next section asks what the always-loser cobidders look like as firms, which disciplines interpretation of what the raw concentration is made of.

5. Economic Content and Ordinary-Loser Alternatives

The validation results show that the award-layer ranking concentrates adjudication-anchored loser-side exposure. The validation target is the broad set of 651 always-loser cobidders—unique always-loser firms that share at least one tender-item with a BEC-active direct CADE defendant—of which 341 (52.4%) satisfy the frequent-loser cut and 310 (47.6%) do not. The label is therefore not frequent-loser-conditioned. This section gives that exposure target economic content through a demanding within-stratum comparison: among always-loser firms, do the firms our screen targets—those observed co-bidding with direct CADE defendants—differ from other always-losers in dimensions relevant for loser-side forensic prioritization? The profile contrast below is run within the frequent-loser stratum (the 341 FL cobidders versus the FL non-cobidders), holding the cut fixed so that the gaps cannot be an artifact of the cut itself. Throughout, "cobidder" denotes an adjudication-anchored exposure label, not legal membership. Because the samples are large (thousands of firms), we report standardized mean differences (SMD, Cohen's d) rather than p-values: significance is uninformative at this N, while standardized differences speak to economic magnitude. The full profile, opportunity-adjusted differences, monotonicity bins, binary-versus-continuous comparison, and the ordinary-loser adjudication table are reported in Appendix D; we summarize the headline findings here.

5.1. What opportunity adjustment removes, and what survives

The headline of this section is deflationary: once we adjust for opportunity—the expected contact and exposure a firm would accumulate given its participation profile—the breadth, persistence, and specialization gaps between cobidders and other frequent losers *vanish*. The item-group concentration difference, raw SMD 0.13, falls to an adjusted SMD of −0.02: the apparent product specialization of cobidders is procurement-environment composition, not an intrinsic portfolio trait. The breadth gap (distinct buyers) and the persistence gap (active years) likewise vanish under opportunity adjustment. The only profile dimensions that survive are the mechanical defendant-contact variables—proximity to the legal anchors attenuates only modestly, the CADE-cases SMD falling from 6.17 to

4.99. But this surviving gap is not independent corroboration: the label is itself defined by defendant contact, so the mechanical share is not separable from any behavioral one. The defendant-contact variables therefore cannot serve as evidence about conduct. Once opportunity is held fixed, almost none of the non-mechanical profile distinction remains (Appendix D).

The raw contrasts below are precisely what opportunity adjustment removes; we report them as descriptive texture, not as findings. The 341 FL cobidders are predominantly electronic-auction firms—a Pregão share of 84.0% against 66.6% for the 2,394 other frequent losers (SMD 0.44)—they reach more distinct buyers (21.9 versus 14.1, SMD 0.53), they are modestly more persistent across active years (SMD 0.39), and they sit closer to the legal anchors used in validation (CADE-cases SMD 6.17). These are breadth and proximity measures, not legal conclusions, and the preceding paragraph has already shown that the breadth and persistence gaps do not survive the opportunity benchmark.

Network specificity cuts in the same deflationary direction. The negative controls show that the raw concentration is generic co-participation arithmetic rather than a signature of the adjudicated network: replacing the true CADE defendants with matched placebo anchors reproduces the raw ranking (real-anchor ROC-AUC 0.761 against a placebo-anchor mean of 0.755; empirical $p = 0.46$), and anchors built from non-CADE high-volume winners generate raw discrimination at least as high (null mean 0.78; $p = 0.91$). Put plainly, non-CADE high-volume winners are ranked at least as tightly to the network as the real defendants. This is exactly what Section 4.2 finds with the opportunity benchmark. The global zero-win definition does retain more raw discrimination than any market-specific alternative (best alternative 0.584 against the global 0.761), so the definition itself is not the weak link (Appendix D).

The most informative diagnostic concerns what the ranking actually orders. Binning firms by participation T_i, cobidder prevalence rises monotonically with participation—from near zero in the lowest bins to roughly 28% in the highest (rank correlation +0.99)—but opportunity-adjusted excess contact $X_i = O_i - E_i$ is flat-to-negative across the same bins (rank correlation −0.25). Higher-ranked firms are not contacting defendants *more than their exposure predicts*; they are simply more exposed. The award-layer rank is therefore an **exposure ranking, not a collusion-intensity ranking**—exactly what the reach-and-limits reframe expects. This also reframes the binary FL14 cutoff as an *administrative* implementation of the exposure ranking, not a structural break: the continuous score $\log(1+T_i)$ ties or beats the FL14 binary in every full-sample configuration (full-universe ROC-AUC 0.761 continuous vs. 0.688 binary), and there is no bunching at the threshold—the density ratio at $T = 14$ is 1.06, essentially flat (Appendix D). FL14 buys a simple reporting cut, not a behavioral discontinuity. This reading is consistent with evidence that auction collusion can operate through bidder roles, repeated relationships, and cartel organization (Baldwin et al., 1997; Porter and Zona, 1999; Pesendorfer, 2000; Asker, 2010).

5.2. Ordinary-loser alternatives and what Section 5 establishes

Could the same profile arise from ordinary, non-collusive reasons that firms lose persistently? The honest summary is that ordinary-loser alternatives are *narrowed but not eliminated*. Cobidders are wider and longer-lived rather than narrow and short-lived, with a lower Convite and minimum-quorum share, which weakens—but does not exclude—the low-capacity quorum-filler story. The proxies that would settle the remaining alternatives—distance-to-winner, geography and logistics, and later wins—are unobserved at the award layer, so benign specialization and geography remain possible (Appendix D). What distinguishes cobidders from these ordinary stories is the one axis the alternatives do not predict: contact with the adjudicated CADE network, the axis the screen is built on.

Section 5 therefore gives the award-layer exposure target modest but bounded economic content, and on balance the limits dominate: most of the raw distinction is procurement-environment composition and partly mechanical proximity that vanishes or attenuates sharply under opportunity adjustment; the breadth and persistence gaps do not survive the adjustment; the raw ranking is reproduced by placebo and non-CADE anchors, so it is not specific to the adjudicated network; the rank is monotone in exposure but flat in opportunity-adjusted excess contact; FL14 is administrative, not structural; and ordinary-loser alternatives are narrowed, not eliminated. The construct is a disciplined exposure ranking whose non-mechanical economic content is modest. It does not establish membership, agreement, liability, or cover bidding in any tender, and it is not diagnostic by itself. Read together with the validation of Section 4 and the operational sequencing of Section 6, it supports triage, not liability: the award layer orders attention toward adjudication-anchored loser-side exposure, and richer records evaluate what the screen has prioritized. The next section turns from content to implementation, asking whether the award-layer rank can be placed before bid-distribution forensics in a sequence that saves bid-microdata recovery while preserving adjudication-anchored recall.

6. From Award-Layer Triage to Bid-Layer Forensics

The preceding sections validate the award-layer ranking on the margin for which it is designed. Section 4 shows that persistent zero-win participation orders adjudication-anchored loser-side exposure; Section 5 shows that the exposed firms have an economically distinctive profile rather than merely high procurement volume. The remaining question is how an enforcement agency should use such a signal. The answer is not to stop at the award layer, and it is not to replace bid-layer forensics. A thin administrative record can rank priorities, but it cannot by itself evaluate the bid conduct that richer cartel evidence requires.

This section turns from validation to sequencing. Whatever value the award-layer ranking retains lies in where it sits: before costly proof-producing work, using records the agency observes broadly—who participated, who won, and who kept appearing without winning—whereas bid-layer forensics use submitted offers, within-tender bid moments, and distance from the winning bid, which are more probative but more expensive to recover, clean, and inspect at scale. The issue is not whether a cheap signal should replace a costly one, but whether it can discipline where the costly one is opened first. We evaluate that architecture in three steps: we place the award-layer ranking and an Imhof–Wallimann-style bid-distribution benchmark on the same adjudication-anchored target, so performance can be read against information cost; we ask whether the two layers are complements; and we implement the operational gatekeeper, in which award records create a survivor pool and bid forensics rank firms inside it. The object is a cost-recall frontier for forensic attention, not a horse race under full observability. This is the paper's central enforcement implication: administrative records rank where proof-producing forensics should begin, while bid-level evidence remains the stage at which conduct is evaluated.

6.1. Bid-Layer Forensics as a Full-Observability Benchmark

The two layers sit at different points in the evidentiary pipeline. The award layer uses records the agency already holds; the bid layer uses the submitted offers, which in the Brazilian setting are not in the routine award file and must be recovered from the underlying `LANCES` bid histories, harmonized, and reduced to within-tender moments before any firm can be scored. The bid-layer benchmark is therefore the forensic stage, not a free comparator: it becomes available only after the agency has paid the cost of opening the bid file. The right question is how much ranking information is already present before that file is opened.

The benchmark is a transparent bid-moment random-forest benchmark inspired by Imhof–Wallimann-style screens rather than a black box. The learner is a random forest (ranger, 500 trees, probability mode, no hyperparameter search), trained on seven within-tender Imhof–Wallimann moments—the coefficient of variation and its within-firm dispersion, skewness, kurtosis, spread, the log min–max ratio, and the second-lowest-bid gap—each averaged across the tenders a firm contests (Abrantes-Metz et al., 2006; Imhof et al., 2018; Imhof, 2019; Huber and Imhof, 2019; Wallimann et al., 2023). The candidate pool is the always-losers for which these moments are computable, the target is the same adjudication-anchored cobidder exposure used throughout Section 4, and all 651 positives—always-loser cobidders—are retained; the small attrition is mechanical and concentrated among non-positives. Timing and bid-revision features are excluded because the records carry no usable per-bid timestamps. The candidate-support audit, the model-design audit, and the leakage-boundary table are reported in full in Appendix E.

Performance.. Table 5 reports precision, recall, and PR-AUC alongside ROC, since the positive base rate is near one percent and ranking quality is what an agency cares about. Two facts stand out. First, on the pooled full-observability diagnostic the cheap award layer is *comparable* to the costly bid benchmark: the award-continuous score reaches ROC 0.760 / PR-AUC 0.143 and the bid-layer random forest reaches ROC 0.717 / PR-AUC 0.116—comparable discrimination at lower informational cost, not a victory of one layer over the other. The bid benchmark is deliberately a transparent, replicable bid-moment comparator rather than a tuned modern learner; a learner with richer tender-level covariates and bid-revision features would plausibly raise the bid-layer ceiling. That would *strengthen* the sequencing argument rather than weaken the paper: a more powerful costly stage raises the value of cheap triage that decides where to deploy it, because the gain from correctly routing scarce forensic attention grows with the payoff of opening the right file. The combined award-plus-bid model reaches ROC 0.756 / PR-AUC 0.188, but it requires bid microdata for every firm and is therefore a full-observability upper bound, not a first-stage rule. Second, the pooled metrics rest on *random* cross-validation folds that are optimistic because the cobidder positives cluster by CADE case. Under case-grouped folds, where the positives from a held-out case never co-train, the bid layer falls hardest: its PR-AUC drops from 0.116 to 0.062 and its ROC from 0.717 to 0.626. Crucially, the *combined* model's PR-AUC also falls—to 0.103, *below* the award-only 0.143—so the apparent complementarity does not survive case-grouped evaluation: it is the label-independent award score, not the bid layer, that continues to rank firms when the case signal thins. Excluding the label-defining tenders leaves the bid ROC essentially unchanged (0.717 $\rightarrow$ 0.713), so the fragility is case clustering, not feature–target overlap.

Complementarity.. The two layers carry partially different ranking information rather than duplicating one another: the Spearman rank correlation between the award and bid scores is only 0.544, and adding the award score to the bid benchmark improves PR-AUC by +0.072 under pooled folds (DeLong $p < 0.001$). The complementarity is conditional on this implemented benchmark and is case-fragile: under case-grouped folds the combined PR-AUC (0.103) falls below award-only (0.143), so what marginal gain exists is mostly the award score *rescuing* the bid layer rather than the reverse. Neither score is proof of an agreement: the bid benchmark measures the shape of a firm's bid distribution. A strict-timing version of the bid benchmark is *blocked*, because within-tender dispersion moments cannot be cleanly restricted to a pre-target window; the award score, by contrast, is label-independent and admits a clean temporal holdout. The full complementarity decomposition, rank-overlap diagram, and leakage boundary appear in Appendix E. What the benchmark establishes is the discriminating ceiling under this implemented full-observability comparator, and how much of it the cheap award layer already reaches at far lower informational cost.

Table 5: Bid-Layer Benchmark and Award-Layer Screen: Ranking Performance

Design	Model	PR-AUC	Prec@500	Rec@500	Prec@1000	ROC
Pooled (random folds)	Award continuous	0.143	0.216	0.166	0.177	0.760
	Award FL14	0.098	0.130	0.100	0.129	0.688
	Bid-layer random forest	0.116	0.180	0.138	0.156	0.717
	Combined award + bid	0.188	0.274	0.210	0.189	0.756
Case-grouped (held-out case)	Award continuous	0.143	0.216	0.166	0.177	0.760
	Award FL14	0.098	0.130	0.100	0.129	0.688
	Bid-layer random forest	0.062	0.086	0.066	0.083	0.626
	Combined award + bid	0.103	0.190	0.146	0.139	0.689
Excl. label-defining tenders	Bid-layer random forest	0.088	0.130	0.114	0.128	0.713

Notes: N = 16,731 always-losers with computable bid moments; 651 in-pool always-loser cobidder positives, all retained. PR-AUC is reported alongside ROC because the positive base rate is near one percent. The **pooled** block is a full-observability diagnostic with random-fold optimism; the **case-grouped** block is the honest robustness test (positives clustered by case never co-train), where the bid layer falls hardest (PR-AUC 0.116→0.062) and the combined model's PR-AUC (0.103) falls *below* award-only (0.143). The award scores are label-independent and so do not change across designs. Cobidders are adjudication-anchored exposure, not cartel membership; false positives are unlabeled firms, not "bad firms." The combined model is a full-observability upper bound, not a first-stage screen, and its complementarity is conditional on this implemented benchmark and case-fragile. Candidate support, model audit, complementarity, and the leakage boundary are in Appendix E.

6.2. *Sequential Gatekeeping and the Cost-Recall Frontier*

Similar discrimination at lower cost does not by itself establish a useful two-stage architecture. The two layers sit at different points in the evidentiary pipeline, and that ordering is the operational lever. The award layer is routine and cheap: it uses records the agency already holds. The bid layer is richer and more probative but costlier: in this setting the submitted offers are not in the award file and must be recovered from the underlying `LANCES` histories, harmonized, and reduced to within-tender moments before any firm can be scored. A model that gives the classifier both layers for every firm—the *joint* model—is a hypothetical full-observability ceiling, never an operating policy: it is an upper bound on what the two layers can do under full observability, but it has already paid the full cost of opening the bid file. The implementable object is therefore sequential: rank where the costly layer should be opened first.

The sequential rule imposes that constraint directly. Firms are ranked using only the award layer; the top K_1 firms form a survivor pool S_{K_1}; bid microdata are recovered *only* for the survivors in S_{K_1}; and the bid-distribution benchmark then reranks firms within that smaller pool by the combined award-plus-bid score, producing the final top-k forensic queue. Full observability is the upper-bound benchmark, not the deployed policy; award-only is the zero-bid-cost baseline that opens no bid records at all; the sequential rule is the operating architecture in between.

Because the opportunity audit shows no robust residual ordering net of exposure, this frontier should be read strictly as the retrospective recovery footprint of a cheap exposure-ranking architecture, not as evidence that the frequent-loser score identifies active conduct. Furthermore, because strict timing for the bid rerank is unavailable with the current bid-layer feature construction, this framework represents an enforcement-accounting baseline conditional on the validated incumbent ranking, rather than a prospective platform-wide deployment test.

What, then, is the frontier accounting? The award ranking orders firms by exposure, and adjudication-anchored cobidders do cluster where exposure is high, mechanically. Routing recovery effort toward high-exposure firms is therefore a coherent evidence-allocation policy even in the absence of any conduct-specific residual: the frontier prices that policy's recovery footprint—how much of the adjudication-anchored target a capacity-constrained agency can reach for a given volume of opened bid records. What the frontier does *not* do is certify that the firms opened first are likelier to have coordinated; Section 4 forecloses that reading.

Cost is not only firms.. The first-order honest point is that the firm count is the wrong denominator for forensic burden. Recovering the within-tender bid moments requires the full tender-item `LANCES` record for every tender a survivor contested, so the relevant cost is measured in opened tender-items and opened bid rows, not in firms. Table 6 reports all three. At the $K_1 = 2{,}000$ operating point the rule opens only 12% of firms (a firm reduction of 88% relative to full observability), but it recovers 34% fewer tender-items and only 33% fewer bid rows. Survivors are the highest-participation always-losers, so opening 12% of firms still opens roughly 67% of the bid rows. The frontier reports firms, tender-items, and bid rows precisely so this divergence is visible: the honest processing-burden cut is about a third at the bid-row level, not the firm-level figure. We report the firm-vs-bid-row divergence (88% versus 33%) as the core caveat, not a footnote.

The frontier, not a cutoff.. Varying K_1 traces a cost-recall frontier rather than identifying a single best rule. At $k = 500$, the number of positives the sequential award→bid rule recovers is non-monotone in K_1: 108 true positives at $K_1 = 500$, 124 at 1,000, 116 at 2,000, 115 at 3,000, and 102 at 5,000 (Fig. 3). Notably $K_1 = 1{,}000$ *beats* $K_1 = 2{,}000$ at $k = 500$ (124 versus 116 true positives, recovering $124/133 = 93\%$ of the joint upper bound's true positives at a 47% bid-row reduction)—further proof that no K_1 is "optimal." The sequential rule never exceeds the joint full-observability upper bound (133 true positives at $k = 500$, 191 at $k = 1{,}000$). $K_1 = 2{,}000$ is therefore *one operating point on the frontier, not a calibrated optimum*: agencies with larger forensic capacity sit higher on the frontier and agencies with less capacity sit lower. The contribution is the frontier itself—the menu of recall-versus-scope trade-offs—not a universal cutoff.

What the bid rerank does, and does not, add.. The bid layer adds little operationally at a fixed survivor pool. The award-defined survivor pool already gathers most of the positives the final queue recovers; the subsequent bid rerank only reorders firms inside that pool rather than surfacing positives the award gate missed. The architecture's value is therefore in cost—preserving much of the target capture while reducing the scope of proof-producing forensics—rather than in the bid layer locating new targets. This is the opposite of a claim that the bid stage drives recovery: it is the award gate that determines which positives are reachable, and the rerank that orders them.

False positives and missed positives.. The frontier is explicit about both error margins. At $K_1 = 2{,}000$, $k = 500$, the final top-500 queue contains 116 true positives and 384 unlabeled firms, for a precision of 0.232 and a recall of 0.178 against the 651 positives;

543 positives are missed overall. We do not read modest precision as failure: in forensic triage, routing a manageable queue to the next, costlier evidentiary stage is the operative property, and the 384 unlabeled firms in the queue are unverified, not exonerated.

Baselines.. The frontier is bracketed by three reference points. Award-only is the *zero-bid-cost* baseline (recall 0.166 at $k = 500$), achievable with no bid recovery whatever. Bid-only and joint scoring are full-observability benchmarks that require opening the entire pool's bid records; joint is the upper bound the sequential rule approaches but never exceeds. A random survivor pool of the same size is far below the award-ranked one, confirming that it is the award ranking, not the mere act of opening K_1 firms, that carries the signal.

The institutional reading follows directly. The rule is not designed to beat full information; it preserves much of the target capture while reducing the proof-producing forensic scope—and even that reduction is honest only at the firm level and modest at the bid-row level. The bid rerank's marginal contribution at a fixed survivor pool is reordering, not new recall. A screen of this kind does not decide liability; it decides, under a stated capacity constraint, which files deserve the next and more expensive form of evidentiary attention.

The frontier is also case-fragile. The single largest CADE case supplies 32.0% of the positives and 45.4% of the true positives in the top-500 queue, so leaving it out lowers sequential recall materially. The signal is real but concentrated; a different case mix would move the frontier, and the operating points should be read as setting-specific rather than portable magnitudes. The estimated ranking is not a portable cartel score; the transferable object is the decomposition framework—label construction, opportunity adjustment, timing discipline, case-composition audit, bid-layer comparison, and cost-recall accounting.

Timing discipline.. The frontier above uses the full feature window, which is close to the adjudication target; the reading caveat stated at the opening of this subsection applies throughout. A stricter timing audit restricts the award-layer score to a 2009–2016 window and evaluates against later adjudication-anchored exposure; this is not a real-time field

Table 6: Cost-Recall Frontier for Sequential Award→Bid Gatekeeping

Rule	K_1	Firms open	Tender items	Bid rows	TP	FP	Recall	Prec.	ΔFirm red.	ΔBid red.	RLoss vs joint
Panel A: forensic queue $k = 500$											
Award-only (no bid)	—	500	41,704	1,356,346	108	392	0.166	0.216	0.97	0.60	0.038
Bid-only (full obs)	—	16,772	116,112	3,393,915	102	398	0.157	0.204	—	—	0.048
Joint (full obs, U.B.)	—	16,772	116,112	3,393,915	133	367	0.204	0.266	—	—	0.00
Sequential	500	500	41,704	1,356,346	108	392	0.166	0.216	0.97	0.60	0.038
Sequential	1,000	1,000	58,408	1,783,750	124	376	0.190	0.248	0.94	0.47	0.014
Sequential	2,000	2,000	77,097	2,283,924	116	384	0.178	0.232	88%	33%	0.026
Sequential	3,000	3,000	87,714	2,575,274	115	385	0.177	0.230	0.82	0.24	0.028
Sequential	5,000	5,000	99,977	2,921,977	102	398	0.157	0.204	0.70	0.14	0.048
Panel B: forensic queue $k = 1{,}000$											
Award-only (no bid)	—	1,000	58,408	1,783,750	177	823	0.272	0.177	0.94	0.47	0.022
Bid-only (full obs)	—	16,772	116,112	3,393,915	157	843	0.241	0.157	—	—	0.052
Joint (full obs, U.B.)	—	16,772	116,112	3,393,915	191	809	0.293	0.191	—	—	0.00
Sequential	1,000	1,000	58,408	1,783,750	177	823	0.272	0.177	0.94	0.47	0.022
Sequential	2,000	2,000	77,097	2,283,924	185	815	0.284	0.185	88%	33%	0.009
Sequential	3,000	3,000	87,714	2,575,274	184	816	0.283	0.184	0.82	0.24	0.011
Sequential	5,000	5,000	99,977	2,921,977	176	824	0.270	0.176	0.70	0.14	0.023

Notes: Evaluation pool: 16,772 always-loser firms with computable complete-Imhof bid-distribution features; positives are 651 always-loser cobidders. The target is *adjudication-anchored exposure*, not cartel membership. "Firms open," "Tender items," and "Bid rows" report the forensic footprint that must be recovered before the rule's final queue can be scored; full observability opens all 16,772 firms / 116,112 tender-items / 3,393,915 bid rows. **The firm count is not the only denominator**: opened tender-items and bid rows better approximate processing burden, and the firm reduction (ΔFirm red.) substantially overstates the bid-row reduction (ΔBid red.) because survivors are the highest-participation firms. **Award-only opens no bid records** as a pure award screen; the bid-footprint columns for the award-only rows are the *hypothetical* scope if it were verified at the bid layer. Bid-only and joint require full bid observability for the entire pool; the sequential rule recovers bid microdata only for the K_1 survivors. Cost reductions are computed relative to full observability. "RLoss vs joint" is recall lost relative to the joint upper bound at the same k. False positives (FP) are non-labeled firms, *not* exonerated. ROC is not the decision metric here; recall against the adjudication-anchored target at a fixed forensic queue size is.

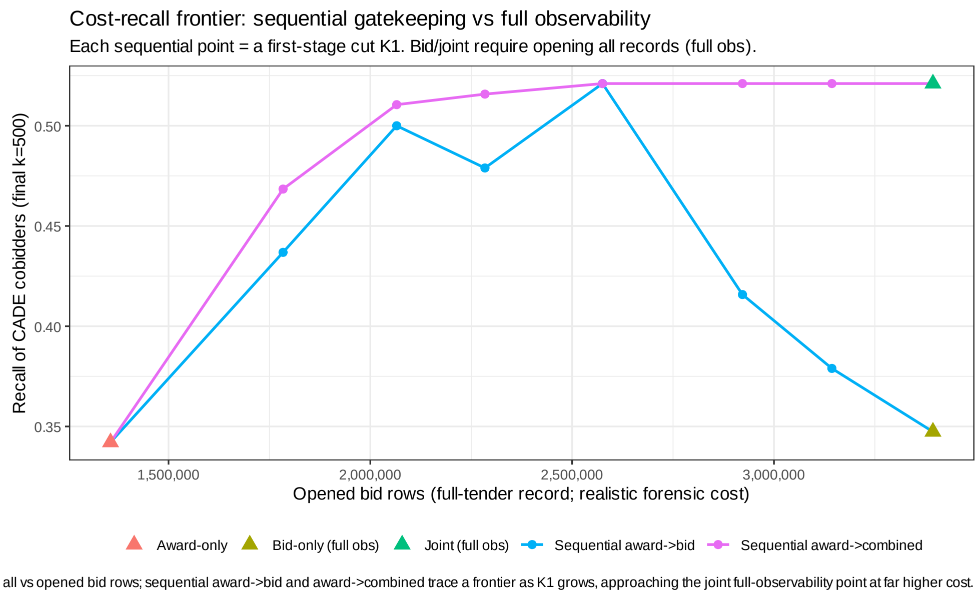

Figure 3: Cost-recall frontier for sequential award→bid gatekeeping. Points trace operational trade-offs in forensic attention, *not* legal proof. The horizontal axis is the recovered bid-record footprint (bid-row reduction relative to full observability) and the vertical axis is recall against adjudication-anchored cobidder exposure at a fixed forensic queue size. Full-observability benchmarks (bid-only, joint) require whole-pool bid recovery and are plotted as horizontal reference lines; the sequential points vary $K_1 \in \{500, 1{,}000, 2{,}000, 3{,}000, 5{,}000\}$ and never exceed the joint upper bound; award-only is the zero-bid-cost baseline. No K_1 is optimal: at $k = 500$ the recovered-positive count peaks at $K_1 = 1{,}000$.

Alt text: A scatter-and-line plot. The horizontal axis is the reduction in recovered bid records relative to opening every firm's bid history; the vertical axis is recall of adjudication-anchored cobidder positives at a fixed forensic queue size. A curve of sequential operating points rises and then falls as K_1 grows from 500 to 5,000, staying below a horizontal dashed line marking the joint full-observability upper bound; it never rises above that line. A second horizontal line marks the lower bid-only full-observability benchmark, and a marker at the far right shows the zero-bid-cost award-only baseline at lower recall. The visual message is that opening fewer bid records costs recall along a frontier, the sequential rule at best matches but never exceeds full information, and the firm-level footprint reduction overstates the bid-row footprint reduction.

deployment, since the labels remain adjudication-based, but it disciplines the information used to rank firms before bid forensics are allocated. Under that restriction the *award-only* score remains usable on the training-AL pool, with ROC 0.684 against the full-window 0.761. The *sequential* rule, by contrast, cannot be cleanly time-limited: the within-tender bid moments that drive the rerank cannot be restricted to a pre-target window without contaminating them with post-window tenders, so a strict-timing sequential frontier is *blocked*. We state this rather than fabricate a number. The conclusion of Section 6 is therefore retrospective and bounded: cheap administrative records can structure the order in which expensive proof-producing evidence is assembled and, on the award layer, survive a timing holdout; the richer bid layer remains the place where conduct is evaluated, and its strict timing properties cannot be certified here.

7. Scope, Limits, and Price Corroboration

Section 6 carries the paper's central enforcement-design result. Price evidence enters here in a narrower, corroborative role only. It is scope evidence: it speaks to whether the award-layer screen points to economically non-neutral environments. It is *not* a damages estimate, an overcharge, a cartel markup, a causal price effect of frequent-loser presence, or proof of a cover-bidding mechanism. Overcharge measurement has its own evidentiary tradition (Bryant and Eckard, 1991; Connor, 2007), distinct from screening (Marshall and Marx, 2012; Harrington, 2008). We state this boundary once and hold to it throughout. The full price regressions are reported in Appendix F.

In the broad item-level regressions, frequent-loser presence is positively associated with log negotiated unit price ($+0.064$, $p = 0.003$; range $+3.6\% - +7.7\%$ across estimators), and the association is larger in pregão than in convite. That broad sign, however, is a weighting result: holding the overlap cells fixed and moving to the within-cell average treatment effect on the treated reverses it (-0.097, $p < 0.001$), because the broad sample mixes a positive between-cell selection component—flagged environments sit in structurally higher-priced cells—with a negative within-cell component. The reversal is robust outside the high-value tail, the direct-CADE overlap estimate is null (-0.061, $p = 0.45$), and the within-cell movement loads on the count of genuine bidders rather than on the frequent-loser count. These two facts—a selection-driven broad imprint and an unidentified within-cell channel—place the evidence on the scope side of the line: it indicates that the award-layer signal points to economically non-neutral environments, no more. The full regressions, the decomposition table, and the bidder-count boundary appear in Appendix F.

7.1. The Instability of a Published Administrative Cutoff

The FL14 cutoff is an administrative implementation, not a structural threshold, and Section 6 already establishes that the ranking carries no conduct-specific residual. The relevant warning is therefore one of institutional design rather than of signal robustness: any *published* administrative threshold invites manipulation once firms learn which observable patterns attract attention. Identity fragmentation and threshold-aware capping are the most corrosive responses—simulated manipulations of this kind can move the rank by several AUC points (Appendix F.6)—and a static disclosed cutoff is the most gameable rule precisely because it can be bunched against. The design answer is not to defend the cutoff but to retire it: use continuous ranks or capacity-constrained top-k queues, refresh the queue as new participation data arrive, monitor bunching below any operational cutoff, and follow award-layer triage with bid-layer forensics whenever bid files can be recovered. The survivor-pool size of such a refreshed queue is itself a point on the cost–recall frontier of Section 6, so investigative capacity and exposure to gaming are chosen together rather than tuned in isolation. None of this rehabilitates the ranking as a liability rule; it is a way to order scarce investigative attention under the limits documented in Sections 6 and 7, and it should never be used as proof of conduct.

8. Conclusion

Cartel enforcement often begins before the agency has the evidence needed to prove a case. Procurement records then pose a problem prior to liability: how should scarce investigative capacity be allocated when the cheap administrative record is broad but legally thin, and the richer bid-level record is costly to recover and evaluate? This paper treats procurement-cartel screening as evidence allocation under costly proof: the contribution is not to collapse the cheap award layer and the costly bid layer, but to sequence them, so that administrative records rank where investigation should begin and bid-level evidence evaluates conduct inside the prioritized set.

The validation evidence maps that division of labor within sharp limits, and establishing those limits is the point. Persistent zero-win participation orders adjudication-anchored loser-side exposure, but a disciplined decomposition—opportunity-cell adjustment, strict rolling-origin timing, leave-one-case-out, and environment-dominance checks—shows that the cheap signal's apparent reach mostly reflects procurement opportunity and case concentration. Once participation volume and opportunity-set exposure are held fixed, the residual ordering is marginal at best and not robust across designs: within comparable opportunity sets the discrimination is near chance, the nested increment is small and only borderline significant, matched-stratum permutation and enrichment tests do not reject the opportunity null, and the performance is concentrated in a single adjudicated case and one procurement environment. Strict prospective ranking fails outside incumbent pools—new entrants cannot be ranked by prior losing, so the ordering is retrospective and incumbent-bound. The ranking does not behave like a generic direct-defendant detector, consistent with its loser-side scope. The general lesson is one of audit discipline: validating an administrative screen against adjudicated cases without adjusting for procurement opportunity systematically over-credits it. Separating this fragile residual from exposure arithmetic and case concentration—and saying plainly where each begins—is the paper's methodological contribution.

The two layers' division of labor is likewise conditional. On the same adjudication-anchored target, the award-layer score is comparable to a transparent bid-moment benchmark, and a combined model adds information only under some validation folds and falls below the award-layer ranking under case-grouped folds; the complementarity is real but case-fragile, not a clean dominance. Joint scoring is a full-observability upper bound; sequential gatekeeping is the architecture relevant once an agency knows where the cheap layer ranks and where it does not, and the cost–recall frontier—not any single cutoff—is the object it offers.

The limits are part of the design. The cheap screen creates forensic priority, not proof.

The validation target is adjudication-anchored exposure, not cartel membership. Price evidence is scope evidence, not damages. The architecture is portable only where award records, repeated participation, comparable procurement categories, and recoverable bid files coexist. A static cutoff can be gamed, so use should rely on refreshed thresholds, continuous ranks or top-k queues, bunching checks, and bid-layer follow-up.

The estimated ranking is not a portable cartel score. The transferable object is the decomposition framework: label construction, opportunity adjustment, timing discipline, case-composition audit, bid-layer comparison, and cost-recall accounting. Cheap suspicion should neither be ignored nor treated as proof. Award-layer triage gives an agency a disciplined way to order scarce investigative attention before the richer evidentiary record is assembled—and, just as usefully, a disciplined way to know when a cheap administrative signal is mostly exposure and concentration rather than evidence. The statistic indicates where legal attention should start, not where legal responsibility ends; mapping that reach, and its edge, is what makes the cheap layer safe to audit with.

References

Abrantes-Metz, R. M., L. M. Froeb, J. F. Geweke, and C. T. Taylor (2006). A variance screen for collusion. *International Journal of Industrial Organization 24*(3), 467–486.

Asker, J. (2010). A study of the internal organization of a bidding cartel. *American Economic Review 100*(3), 724–762.

Athey, S., J. Levin, and E. Seira (2011). Comparing open and sealed bid auctions: Evidence from timber auctions. *Quarterly Journal of Economics 126*(1), 207–257.

Bajari, P. and L. Ye (2003). Deciding between competition and collusion. *Review of Economics and Statistics 85*(4), 971–989.

Baker, J. B. (2003). The case for antitrust enforcement. *Journal of Economic Perspectives 17*(4), 27–50.

Baldwin, L. H., R. C. Marshall, and J.-F. Richard (1997). Bidder collusion at forest service timber sales. *Journal of Political Economy 105*(4), 657–699.

Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy 76*(2), 169–217.

Bryant, P. G. and E. W. Eckard (1991). A price-fixing probability index. *Review of Economics and Statistics 73*(4), 749–752.

Chassang, S., K. Kawai, J. Nakabayashi, and J. Ortner (2022). Robust screens for noncompetitive bidding in procurement auctions. *Econometrica 90*(1), 315–346.

Conley, T. G. and F. Decarolis (2016). Detecting bidder groups in collusive auctions. *American Economic Journal: Microeconomics 8*(2), 1–38.

Connor, J. M. (2007). Price-fixing overcharges: Legal and economic evidence. In *Research in Law and Economics*, Volume 22, pp. 59–153. Elsevier.

Coviello, D. and S. Gagliarducci (2017). Tenure in office and public procurement. *American Economic Journal: Economic Policy 9*(3), 59–105.

Decarolis, F. (2014). Awarding price, contract performance, and bids screening: Evidence from procurement auctions. *American Economic Journal: Applied Economics 6*(1), 108–132.

Decarolis, F., R. Fisman, P. Pinotti, and S. Vannutelli (2025). Rules, discretion, and corruption in procurement: Evidence from italian government contracting. *Journal of Political Economy Microeconomics 3*(2), 213–254.

Decarolis, F., L. M. Giuffrida, E. Iossa, V. Mollisi, and G. Spagnolo (2020). Bureaucratic competence and procurement outcomes. *Journal of Law, Economics, and Organization 36*(3), 537–597.

DeLong, E. R., D. M. DeLong, and D. L. Clarke-Pearson (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics 44*(3), 837–845.

Fazekas, M. and G. Kocsis (2020). Uncovering high-level corruption: Cross-national objective corruption risk indicators using public procurement data. *British Journal of Political Science 50*(1), 155–164. Verified 2026-06-04: DOI 10.1017/S0007123417000461.

Harrington, J. E. (2008). Detecting cartels. In P. Buccirossi (Ed.), *Handbook of Antitrust Economics*, pp. 213–258. MIT Press.

Hovenkamp, H. (2005). *Federal Antitrust Policy: The Law of Competition and Its Practice.* Thomson West.

Huber, M. and D. Imhof (2019). Machine learning with screens for detecting bid-rigging cartels. *International Journal of Industrial Organization 65*, 277–301.

Imhof, D. (2019). Detecting bid-rigging cartels with descriptive statistics. *Journal of Competition*

Law & Economics 15(4), 427–467.

Imhof, D., Y. Karagök, and S. Rutz (2018). Screening for bid rigging—does it work? *Journal of Competition Law & Economics 14*(2), 235–261.

Kawai, K. and J. Nakabayashi (2022). Detecting large-scale collusion in procurement auctions. *Journal of Political Economy 130*(5), 1364–1411.

Marshall, R. C. and L. M. Marx (2012). *The Economics of Collusion: Cartels and Bidding Rings*. Cambridge, MA: MIT Press.

OECD (2022). Data screening tools for competition investigations. Technical Report 284, OECD Publishing, Paris.

Pesendorfer, M. (2000). A study of collusion in first-price auctions. *Review of Economic Studies 67*(3), 381–411.

Porter, R. H. and J. D. Zona (1993). Detection of bid rigging in procurement auctions. *Journal of Political Economy 101*(3), 518–538.

Porter, R. H. and J. D. Zona (1999). Ohio school milk markets: An analysis of bidding. *RAND Journal of Economics 30*(2), 263–288.

Sánchez Graells, A. (2019). "Screening for Cartels" in Public Procurement: Cheating at Solitaire to Sell Fool's Gold? *Journal of European Competition Law & Practice 10*(4), 199–211.

Stigler, G. J. (1970). The optimum enforcement of laws. *Journal of Political Economy 78*(3), 526–536.

Wallimann, H., D. Imhof, and M. Huber (2023). A machine learning approach for flagging incomplete bid-rigging cartels. *Computational Economics 62*(4), 1669–1720.

Wils, W. P. J. (2007). Leniency in antitrust enforcement: Theory and practice. *World Competition 30*(1), 25–64.

Wils, W. P. J. (2016). The use of leniency in EU cartel enforcement: An assessment after twenty years. *World Competition 39*(3), 327–388.